DATA ENGINEERING TECHNICAL ASSIGNMENT

GSS survey pipeline report

Religiosity and media use, 2014 to 2018: how the data was built, what it shows and how far to trust it.

7,753
distinct interviews, each counted once

2,867
held by both cuts, matching field for field

44.7%
never read a newspaper in 2018, up from 30.3% in 2014

2.74
TV hours a day in 2018, against 2.85 in 2014: no clear change

In short. Two overlapping NORC extracts load into DuckDB, and dbt turns them into one table of interviews. One command rebuilds everything; it publishes only if every check passes, and otherwise the last good run stays in place. The clearest finding is the rise in never-readers of newspapers. TV viewing shows no clear link to job satisfaction. Estimates use the WTSSPS survey weight unless marked otherwise, with raw counts alongside.

1. The extracts and why these years

	Survey years	Interviews	Shared with the other cut
Cut A	2014, 2016	5,405	2,867 (53.04%)
Cut B	2016, 2018	5,215	2,867 (54.98%)
Union	2014, 2016, 2018	7,753	Limit is 10,000 distinct

Each cut is near 5,000 records and holds a year the other lacks. Sharing a whole wave is the natural overlap, and it tests whether two independent exports agree: all shared interviews match field for field. I chose 2014 to 2018 because every concept is observed under the same definitions; 2012 gave fewer web-duration answers, and 2021 changed survey mode and wording. The cost is recency: these results describe those years, not today.

2. Which variable answers each question

Question	Variable	Why, and how non-answers are handled
Working status	WRKSTAT	Code 3 (job, temporarily absent) counts as working; Appendix C tests moving those 150 people.
If working, over 12 months in lifetime?	EVWORK	**Cannot be answered for workers.** The item skips workers and job seekers, and asks about at least one year. Reported for nonworkers with bounds.
Education level	DEGREE, EDUC	DEGREE is the credential and is used for grouping. EDUC gives years of schooling.
Industry	INDUS10	Nonworkers can report an earlier job, so current industry needs the working filter.
Satisfaction with work	SATJOB	2016 ballot A, form Y is marked not asked. Job seekers and homemakers also answer it, so Q4 filters to workers.
Life after death	POSTLIFE	Asked on every ballot. Don't know and no answer are counted separately.
Prayer	PRAY	Ordered frequency; 99% valid.
Religion raised in	RELIG16	Upbringing, kept apart from current RELIG.
Newspaper	NEWS	Ballots A and B only, about two thirds of each wave. Ballot C is marked not asked.
TV	TVHOURS	Same ballots. It measures hours per day, not how often someone watches.
Interest in technology	INTTECH	Ballot and form vary by year; about half of each wave. It measures interest, not use.

Also carried: RELIG, ATTEND, RELITEN, WWWHR, WWWMIN, AGE, SEX, REGION, BALLOT, FORM, MODE and both weights.

3. Data profile

Written before any pipeline code, in `docs/profile.md`, with missing shares for every variable and year in `docs/profile.csv`. Interviews per wave: 2,538 in 2014, 2,867 in 2016 and 2,348 in 2018. The media items only cover ballots A and B, which GSS assigns at random, and the routed items only describe the people they were routed to. No selected concept is missing a whole wave. Valid answers as a share of all interviews in 2014, 2016 and 2018: NEWS 66% in each; TVHOURS 66% in each; WWWHR 55%, 55%, 58%; INTTECH 49%, 48%, 50%.

In the export	Stored status	Meaning
.d	dont_know	Asked, and the respondent did not know.
.n	no_answer	Asked, and no answer was recorded.
.i	inapplicable	Skipped by routing, such as a worker on EVWORK.
.i outside a reviewed ballot or form	not_asked_ballot	This ballot never had the question. The original code stays in source_status.
.y, or a field missing from a file	not_asked_year, absent_field	Not asked that year, or left out of the extract. Neither is a survey answer.

4. Database and data model

DuckDB gives analytical SQL and dbt support in one local file, with no server or cost for a reviewer. The model has one key: **(survey year, respondent ID)**. ID alone would merge 4,886 separate interviews because IDs repeat each wave.

Schema	Object	One row per	Stored	Why, and what changes at full scale
raw	records, files	file + row	Table	Exactly what was received. At scale: columnar files by wave.
staging	stg_membership, stg_answers	file + row; interview + field	View	Which file holds each interview, and the JSON unpacked. Read once per build, so they stay views.
marts	answers	interview + variable	Table	Decodes every item once. At scale: incremental by wave, with only the needed variables.
marts	respondents	interview	Table	What analysts query. Saves re-pivoting on every read. At scale: stays a table, with fewer columns.
metadata	seeds	code, variable or rule	Seed	Versioned meaning of each code, and where it is valid.

Status lives on `marts.answers`, so no missing type can pass for an answer; a null in the wide table means no genuine answer, and `answers.status` says why. To add a variable, an analyst re-exports, reviews the new hash in the manifest and adds rows to two seeds; no model changes. The naming convention is in `docs/naming.md`.

Weight. WTSSPS is primary, because NORC's FAQ names it for most analyses and it keeps continuity. NORC's report MR135 recommends WTSSNRPS for cross sections, so CSV 10 runs the headline estimates under both: none moves by more than 0.6 points and no Q3 leader changes.

5. Libraries, and what I did not use

Used	For	Not used	Why not
DuckDB	Storage and SQL	Postgres, SQLite	A server to run, or weak analytics. Neither helps at this size.
dbt Core	Models, tests, docs	dbt packages	A few singular tests were clearer than another dependency.
Python standard library, SciPy	Loading and checks; Q4 rank tests	pandas, Airflow, Dagster	SQL handles the aggregation, and a scheduler for a manual two-file batch adds more than it saves.
Hash-locked virtual env	Reproducible install	Docker	A second install path to maintain; the lock file already pins every wheel by hash. I would add it for a shared server.
ReportLab	This report	Terraform	Nothing is deployed; `docs/production.md` sketches AWS.

6. Results, and how far to trust them

Q2 and Q6 are reported by wave. Q1, Q3, Q4 and Q5 ask about people rather than change, so they pool the three waves. WTSSPS sums to the interviews in each wave, so each wave counts by its sample size.

Q1. Prayer by religion raised in (weighted share of each group)

Trust: good, large groups

Raised in	n	Several times a day	Once a day	Several times a week	Once a week	Less than once a week	Never
Protestant	4,007	34.2%	29.4%	10.2%	6.0%	8.5%	11.6%
Catholic	2,515	22.9%	32.7%	10.6%	7.7%	11.1%	14.9%
None	682	14.7%	18.0%	9.9%	4.9%	13.3%	39.2%
Jewish	137	17.2%	10.7%	5.1%	5.2%	20.9%	40.9%
Christian	111	19.1%	29.8%	12.6%	6.1%	13.0%	19.5%

Among those raised with no religion, 39.2% never pray. Raised Muslim, 61.0% (n=49) pray several times a day. **Caveat:** Buddhism, Hinduism, Muslim/Islam, Native American, Orthodox-Christian, Other each have under 50 people and are flagged in CSV 01; read them as indications only.

Q2. Belief in life after death

Trust: high

Yes among valid answers: 79.1% in 2014, 79.8% in 2016, 80.6% in 2018. Not asked: 0 respondents in any wave. Don't know: 291, 271 and 216, excluded and not treated as no. The wave differences are small.

Q3. Most common industry among current workers, by degree

Trust: medium

Degree	Leading industry	Share	n	Runner-up	Share	z
Associate/junior college	Hospitals	8.5%	41 of 398	Restaurants and other food services	4.6%	1.81
Bachelor's	Elementary and secondary schools	9.8%	109 of 1,037	Hospitals	6.3%	2.30
Graduate	Elementary and secondary schools	13.9%	94 of 582	Colleges and universities, including junior colleges	11.7%	0.85
High school	Construction	9.0%	186 of 2,275	Restaurants and other food services	7.6%	1.33
Less than high school	Construction	15.0%	53 of 384	Restaurants and other food services	12.1%	0.89

Caveat: no industry holds a majority, so each leader is only the largest of many fine-grained labels. Column z is the lead over the runner-up in standard errors: only Bachelor's clears 2, and the other 4 could reverse in another sample.

Q4. TV hours and job satisfaction

Trust: good, no large link

Current workers	n	Effective n	TV hours/day, weighted	Unweighted
Very satisfied	1,415	988	2.31	2.32
Moderately satisfied	1,091	717	2.30	2.37
A little dissatisfied	251	156	2.22	2.39
Very dissatisfied	88	58	2.24	2.55

No visible relationship. Weighted means differ by at most 0.09 hours among the 2,845 of 4,712 workers with both answers, and the direction is not stable: the unweighted figures rise from 2.32 to 2.55 as satisfaction falls, while the weighted ones fall from 2.31 to 2.24. Median TV is 2 hours in every group. Unweighted rank tests agree (Spearman rho -0.011, p=0.57; Kruskal-Wallis p=0.44). **Caveat:** this cannot rule out a small link, the very dissatisfied group is small, and it shows association only.

Q5. Nonworkers who ever worked for a year

Trust: good, for those asked

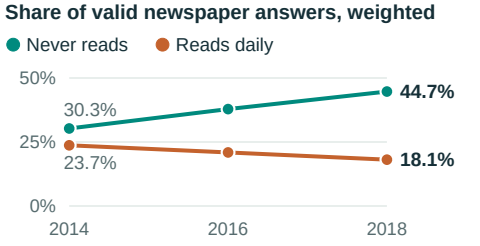
87.5% of the 2,726 who answered (90.5% unweighted). The 306 job seekers were never asked, and 2 answers are missing. Across all 3,034 nonworkers the possible range is 78.5% to 88.8%. **Mitigation:** none of the 259 who said no has an industry on record, while 2,447 of the 2,467 who said yes and 291 of the 306 job seekers do. Counting those job seekers as having worked narrows the range to 88.3% to 88.8%. **Caveat:** the narrowed range assumes a recorded industry means at least a year of work, which the data cannot prove; only the wider range is free of that assumption.

Q6. Newspaper reading and TV from wave to wave

Trust: medium

Wave	Never reads	Change	z	Reads daily	Change	z	TV hours	Change	z
2014	30.3%	-	-	23.7%	-	-	2.85	-	-
2016	37.9%	7.57	4.01	20.9%	-2.79	-1.68	2.78	-0.07	-0.73
2018	44.7%	6.85	3.24	18.1%	-2.81	-1.66	2.74	-0.03	-0.28

Newspaper reading fell, TV did not clearly change. Never-readers grew by 7.57 and 6.85 points, 3 to 4 standard errors each time. Daily reading fell 2.79 and 2.81 points. Neither step clears 2 standard errors alone, but the four-year drop of 5.6 points does (z -3.22). **Caveat:** these are repeated cross sections and not the same people, and the errors allow for weights but not for strata or clusters. Two steps cannot establish a long-run trend.



Overlap audit: is anyone counted twice?

Cut A holds 5,405, cut B 5,215, both 2,867 (53.04% of A, 54.98% of B), and 7,753 are distinct overall. The analysis table has 7,753 rows and 0 duplicates. Every result above reads that table.

7. Efficiency

Query plan. The slowest query, `14_work_status_sensitivity`, took 104 ms when `respondents` was a view and 19 ms as a table, because it no longer re-pivots 232,590 answer rows. The table costs 141 ms to build and repays that in 2 reads per rebuild. Appendix A has the plan. **A new extract:** everything rebuilds, in about 15 seconds here, with no partial state. A changed hash or new year must be reviewed first. **Deliberately not built:** incremental models, partitioning, a scheduler, containers and cloud resources. None pays for itself at this size.

8. Running it, and what happens when it fails

Order: `source/run.py` validates and loads, runs dbt build (seeds, models, tests), generates docs, runs the analyses, then publishes. **Trigger:** a person after reviewing a new extract, and CI on each push; a failed CI run flags the commit but does not block a merge. **Failure:** any failed step stops the rest, including a failed test, which blocks docs and results. Only a final pointer swap publishes a run, so a rerun is always safe. **Evidence:** `build/run.log` names each stage and any failed test, and each run's manifest records hashes of inputs, definitions and outputs. **Configuration:** no credentials exist; the only setting is the optional `GSS_BUILD`. **Tested on** Windows 11 with Python 3.12.

Bad input stops the whole run and names the problem. Appendix B lists 19 corruptions run with `source/failure_check.py`. Loading twice or in either order gives identical data.

9. Risk as data and sources grow

Rows x columns	Answer rows	JSON size	Load	dbt build	Result
75,700 x 30	2.3 M	37 MB	2.3 s	11.4 s	0.08 GB database
75,700 x 100	7.6 M	97 MB	6.1 s	51.6 s	0.14 GB database
75,700 x 300	22.7 M	269 MB	16.7 s	189.7 s	**Out of memory and temporary disk** building marts.answers

One Windows machine (16 logical cores, 32 GB), synthetic rows copied from real records: `source/scale_check.py`.

What breaks first: the dbt build of `marts.answers`. It handled 100 variables and failed at 300, when memory filled and its temporary spill then used up the disk, which is only 4% of the variables in the full file. It crosses every interview with every variable. **What grows fastest:** that same table, respondents times variables, which is about 530 million rows for the full file. **What I would change, in order:** load only the variables needed (about 100 work today), hold typed columns in staging instead of JSON, build by wave, and store only non-skipped answers. With more sources, manual review of each code and routing rule becomes the real bottleneck.

10. Limitations and what remains unsolved

Limitation	What I did	Still open
EVWORK skips workers and job seekers	Bounds, narrowed using recorded industry	Work history of current workers is unknowable from these extracts
No strata or cluster fields: I did not select VSTRAT or VPSU	Weighted estimates with Kish effective n	True design-based errors need a new export (section 13)
Misaligned columns upstream	Overlap comparison covers 2016 only	A shift inside a single-cut wave is not detectable (Appendix B)

11. Trade-offs I chose

Choice	What it gains	What it costs
Full rebuild on each run	No partial state, and a simple recovery story	Wasteful at scale
Stop on any bad input	Nothing that fails a check is published	One bad row blocks the whole run
Built and checked for these two extracts and three waves	Exact checks and a simple design that fit the assignment	A different input needs review and new expected counts first
Long answers table plus a wide table	Add variables with no model change, and easy analyst queries	A dense table that grows quickly

12. What I missed first, and how I fixed it

AI reviews of my earlier versions (section 15) found these. Each fix is covered by a test or the evidence files.

What was wrong	Why it mattered	Fix
Required answers were unweighted, although I had named a primary weight	The Q6 and Q4 conclusions were partly artifacts	Weighted columns on Q1 to Q6, with Q4 and Q6 rewritten
A truncated or empty wave could publish	Half of 2018 could vanish with every check green	Per-wave row counts and coverage floors as dbt tests
Routing gaps	Impossible answers could pass	SATJOB, web-time and unknown-ballot rules
Renaming a cut made the audit report 0	A silent wrong answer	Cuts named in a seed; a rename now fails a test
Nothing checked the weights behind the headlines	One bad weight could swing a published figure	Each wave's WTSSPS must sum to its interviews
The Q5 range ignored evidence in the data	A ten-point range that could be narrowed	A labelled narrowing beside the strict bounds
dbt was flat, with queries outside it	No layers, lineage or seed documentation	Staging unpacks, marts decode; analyses use ref(); seeds documented

13. Future considerations

The first five go beyond the two chosen extracts, so I did not build them; section 9 covers scale. The last three are problems a final AI review found, which I left unchanged.

Item	How I would do it
True survey errors	Re-export with VSTRAT and VPSU, review the new manifest, replace Kish z with design-based errors, and weight the Q4 rank tests.
Subtler bad values	Reviewed per-wave bands for category shares, a weight cap, and row-level quarantine for agreed cases. Today's checks miss a wave of weights all set to 1, or a loss under about 20%.
Deeper tests	A truth table for the `work_status` seed, and fixtures for the remaining queries and standard errors.
Other machines	Linux and macOS commands in the README, and a hash lock for the report renderer.
Production and sources	The AWS design in `docs/production.md`. Pew's Religious Landscape Study as a survey-level check, never joined at person level; Yaqeen format data only in aggregate, with permission and no religious identifiers.
Wrong: CSV 14 column	Q5 bound rows show a denominator of 2,726 but divide by all nonworkers; the estimates are right. Report the real divisor and rerun the benchmark.
Wrong: memory claim	Section 9 says memory filled at 300 variables; the saved evidence has only the disk error. Rerun the scale check recording memory and free disk.
Wrong: Q2 label	Marked high trust, yet about a tenth of each wave said don't know. Add a range counting them as no or yes, as in Q5, plus standard errors, then review the label.

14. Assumptions and divergence from the design note

Assumptions: equal-length columns align by position in the export; code meanings hold across these waves; ID is unique within a wave; NORC's released counts are correct. **Divergence:** the design planned to insert new files into one database; the build instead rebuilds a fresh candidate and publishes atomically, which makes recovery trivial. See `docs/design.md`.

15. AI use and human oversight

Tools: Codex and Claude Code, for writing the code and report, testing and review. **Extent:** most code and much of the prose was AI-written, so "I" here often means work I directed and did not type. **My part:** I chose the extracts and years, supplied both cuts, challenged the TV and employment interpretations, and decided which review findings to act on. **Time:** roughly six hours in total.

Appendix A. Query plan for the most expensive query

Query: `14_work_status_sensitivity.sql`. One thread, the median of three warm runs. The full plans are in `report/query-plans.txt`.

	Before: respondents as a view	After: respondents as a table
Plan	Scan 232,590 rows of answers, hash-group them into one row per interview, hash-join work status, then run the query	Scan 7,753 rows of respondents, then run the query
Read time	104.3 ms	19.4 ms
Extra cost	None	141 ms build, 512 KiB storage
Result rows	24	24 (checked equal)

Timings vary by machine. At full scale the view would re-pivot hundreds of millions of rows on every read, so the difference would grow with the data.

Appendix B. What happens to bad input

Each row corrupts a fresh copy of the project and runs the real pipeline. The previous published run survived every failure. Reproduce with `python source/failure_check.py`.

What goes wrong	Stopped by
Truncated file	loader
File edited without review	loader
Duplicate interview in a file	loader
Overlap answers disagree between cuts	loader
Column with no catalog entry	loader
Label not in the codebook	invalid_answers
TV hours out of range (99)	invalid_answers
A wave outside the reviewed years	invalid_answers
Half of 2018 deleted, manifest resealed	wave_row_counts
One 2018 weight set to 5000	wave_weight_totals
POSTLIFE blank for all of 2018	wave_coverage
PRAY column missing from cut B	wave_coverage
EVWORK answered by a full-time worker	survey_routing
NEWS answered on ballot C	survey_routing
SATJOB answered by a retiree	survey_routing
Web hours from someone who said no web	survey_routing
A ballot the rules do not cover	survey_routing
Cut renamed but seed not updated	metadata_contract
Column shifted one row within a wave	**Not detected** (known limit, section 10)

Appendix C. Sensitivity of Q5 to temporarily absent workers

Counting the 150 people with a job who were temporarily absent as not working moves the weighted Q5 range from 78.5%-88.8% to 74.8%-89.3%. The Q3 leaders do not change. See `report/results/14_work_status_sensitivity.csv`, which also has the unweighted figures.

Source

Davern, Michael; Bautista, Rene; Freese, Jeremy; Herd, Pamela; and Morgan, Stephen L.; General Social Surveys, 1972-2024 [machine-readable data file]. Principal Investigator, Smith, Tom W.; Co-Principal Investigators, Rene Bautista, Jeremy Freese, Pamela Herd, and Stephen L. Morgan; Sponsored by National Science Foundation. NORC ed. Chicago: NORC, 2025: NORC at the University of Chicago [producer and distributor]. Data accessed from the GSS Data Explorer at gssdataexplorer.norc.org.